

Biblical Flood and geological deluge: the amicable dissociation of geology and Genesis

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Abstract: This paper summarizes debates, among European geologists in the early nineteenth century, about the possible equivalence (or non-equivalence) between the biblical account of Noah's Flood, and new and cumulative evidence for an exceptional watery catastrophe or 'geological deluge' in very recent Earth history. The 'diluvial theory' deserves to be taken seriously as an attempted explanation of some extremely puzzling physical features (many of them reinterpreted later as traces of a glacial 'catastrophe' or Ice Age). The 'geological deluge' was eventually recognized as having been far earlier in Earth history than any event recorded by literate human societies. Among geologists, although not always among the wider public, this gradual dissociation between biblical Flood and geological deluge was generally amicable, not acrimonious. It was facilitated by the concurrent development of biblical scholarship, which showed that earlier literalistic interpretations were no longer tenable (and were also destructive of religious meaning). What was transposed into geology in the course of these debates was the strong Judaeo-Christian sense that the world has had a directional and contingent history, which might have been punctuated by occasional catastrophic events.

Historical work on the relations between any of the natural sciences and any of the world's religions should always specify clearly the period, the place and the persons that are under discussion, and also define which social groups were involved. Historians of the sciences now rightly reject attempts to impose any single or simple pattern of, for example, either endemic conflict or perennial harmony, because these are incompatible with the variety and complexity of the historical evidence (see, e.g. Brooke 1991).

This paper is about the 'when' of the late eighteenth and early nineteenth centuries; the 'where' of the whole of Europe (including its offshore islands of Britain and Ireland); and the 'who' of the naturalists who at that time were beginning to call themselves 'geologists'. These are the parameters that Karl von Zittel (1839–1904) used when, over a century ago, he famously defined the decades around 1800 as 'das heroische Zeitalter der Geologie' (von Zittel 1899, p. 76). He too focused on research by Europeans, because during geology's 'heroic period' North America and the rest of the world were still marginal to high-level scientific debate (although treated as valuable sources of new observations and specimens). And he too concentrated on the social group of the leading scientific figures, and on what they were doing and discussing among themselves, rather than on the dissemination and reception of their ideas among the wider public (the latter is a very different story: for the British case, see O'Connor 2007b).

Within these parameters, this paper focuses on one celebrated (or notorious) case of the relation between geology and religion, namely the attempts to interpret certain physical features as traces of the Flood recorded in the book of Genesis, and conversely the attempts to reject any such correlation.

Earth's history, and its timescale

In late eighteenth- and early nineteenth-century Europe, during what political historians often call the Age of Revolution and the subsequent Age of Reform, the Earth sciences were radically transformed by becoming deeply historical in their outlook and practice. Those who pursued these sciences came to recognize not only that the Earth as a whole has had its own history, but also that its features (particular mountains, volcanoes, rocks, fossils and so on) likewise have specific histories built into them, and cannot be understood solely in terms of unchanging ahistorical 'laws of nature'. This radically new outlook on the natural world was the result of a deliberate transposition of methods and concepts from the writing of human history into the study of the Earth, to reconstruct the Earth's own history (in modern terms, geohistory) in all its complex particularity (Rudwick 2005, 2008).

This geohistorical research strategy was so successful and productive that most modern geologists are unaware of it. They use it routinely and take it

completely for granted; they ignore the fact that it was achieved at a specific time and place in human history. But it was not achieved against resistance from a reified entity called 'religion' or from a monolithic entity called 'the Church'. On the contrary, it was positively facilitated by the strongly historical orientation of the Christian religion: the strong sense of an unrepeatable directional movement in human history, not deterministic or predictable but always contingent. In traditional terms, this history, both terrestrial and cosmic, stretched from creation through the incarnation of Christ towards the end of the world (hence the traditional timescale of years BC and AD, now globalized as years BCE and CE). This fundamental concept of contingent directional change could be, and in the event was, extended from recorded human history back into what turned out to be the depths of prehuman geohistory.

Contrary to the assertions of some modern crusading atheists, the possibility of a greatly extended timescale for geohistory was not a religious problem among naturalists (it sometimes was among the wider public, particularly in Britain: O'Connor 2007a). In fact, the idea of an inconceivably lengthy past was, in itself, far from novel. The traditional short timescale of a few millennia for the whole of cosmic history (James Ussher's notorious 4004 BC date for creation was just one of many competing alternatives) had for centuries been juxtaposed to the equally traditional alternative of eternalism, according to which the timescale of the universe was unimaginably vast because it was infinite (Rudwick 1986). It was only during the eighteenth century that a third alternative, transcending this ancient polarity between two equally un-modern options, began to emerge. This was the possibility that the timescale of the Earth's history, and that of the universe as a whole, might be unimaginably lengthy yet not infinite. For the first time, this made it possible to conceive that the Earth might indeed have had a history that was reliably knowable, even though most of it apparently predated any human beings who might have recorded it, rather than just the endless recycling of similar events from and to eternity (Rudwick 2005, pp. 115–130).

By the 1780s at the latest, all knowledgeable naturalists with interests in the Earth sciences had in effect adopted this third (and modern) concept of the Earth's timescale. Although they had no way of quantifying it, they were tacitly agreed that it must be inconceivably vast in relation to human lives or even to the totality of recorded human history. At the same time, however, most of them claimed that through these vast spans of time the Earth had developed directionally towards its present state, that humans were relatively recent newcomers, and that the Earth had therefore not been the same kind

of place from all eternity. To put it another way, they firmly rejected any suggestion that the whole of geohistory was confined to a few millennia; but at the same time most of them also rejected any suggestion that humans had always been around, let alone that the timescale might be infinite because the Earth was eternal. In effect, they rejected anything analogous either to the 'young-Earth' beliefs of modern creationists or to the 'steady-state' theorizing (albeit currently out of favour) of some modern cosmologists.

What eighteenth-century naturalists found most persuasive, as evidence for a lengthy but not eternal geohistory, was a product of their fieldwork in many parts of Europe. This was the discovery that the pile of stratified 'secondary' rock formations, the *monti secundari* or *Flötzgebirge*, was immensely thick, at least in some regions. Furthermore, it seemed almost incontrovertible that many of these strata must have accumulated extremely slowly and in very tranquil conditions. An outstanding example was the Plattenkalk or 'lithographic stone' of Solnhofen in Bavaria, with its exceptionally well-preserved fossils. Even if the underlying 'primary' or basement rocks, the *monti primari* or *Urgebirge*, were attributed to the very origin of the planet, it became clear that the total timescale of subsequent geohistory must be unimaginably vast in relation to the whole of recorded human history (Rudwick 2005, pp. 84–98, 115–130).

This inference was surprising and unexpected, but it was no problem for the many leading naturalists who regarded themselves as religious believers. They saw no conflict between a lengthy geohistory and the two creation stories in Genesis (Chapters 1–3), because they were aware that biblical scholars, since the earliest history of the Church, had drawn attention to the limitations of biblical literalism. For example, it had often been pointed out that the seven 'days' of the first creation story could hardly denote periods of 24 hours (the first three were described as preceding the origin of the Sun itself). Each 'day' was better regarded as a time of special divine significance, rather like the 'day of the Lord' in prophetic discourse. The theological significance of the story, and hence its practical religious meaning, lay not in any quantified chronology but rather in its reiterated claim that the world was not self-generated or eternal, but the product of divine intention directed towards what was good ('And God said, let there be ... and it was so; and God saw that it was good').

The biblical Flood in geohistory

However, the story of Noah's Flood, later in the Genesis text (Chapters 6–8), posed very different

problems. Unlike the creation stories, the biblical Flood story claimed to record events within human history, not before it. Therefore it was reasonable to suppose that it would be possible to locate it on the human-historical timescale of years BC, by searching all known textual records, both biblical and non-biblical, using the scholarly methods of the well-established historical science of 'chronology' (Grafton 1991). Also, because the Flood was recorded as having been drastic in its physical effects, as well as catastrophic for humankind, it was also reasonable to expect to find natural evidence for its historical reality, in addition to textual evidence. Naturalists in a much earlier period, in the time of Niels Stensen [Steno] (1638–1686) and John Woodward (1665–1728), had in effect attributed the whole pile of 'secondary' formations to the Flood, although in consequence they had had to interpret the putative event in far from literalistic ways. However, more extensive fieldwork over the next hundred years, leading to the discovery of the sheer thickness of the pile, had made this interpretation less and less plausible. By the late eighteenth century, the only materials still attributed to the biblical Flood, and therefore sometimes termed 'diluvial', were the so-called 'superficial' deposits such as boulder clay (till) and coarse gravels, the *ausgeschwemmte* materials or *terrains d'atterrissement*, overlying all the 'regular' or clearly stratified secondary formations.

It followed that the physical event that had formed the superficial deposits must have taken place (relatively) very late in geohistory. Thus it was reasonable to conclude that it might be recent enough to be the trace of the biblical Flood itself. If so, the diluvial event formed a unique boundary between the (relatively) very brief span of human history and the vast spans of prehuman geohistory; or rather, a unique zone of overlap between the two. Naturalists, including those who were religious, tried hard to explain the diluvial event in purely natural terms, by enlarging the scale of what the Anglo-Genevan savant Jean-André de Luc (1727–1817) termed 'causes actuelles' (actual causes); that is, processes visibly operating now ('actually' in the older sense of the word) in the present world. De Luc himself, for example, conjectured in 1779 that the 'diluvial' features were due to the sudden collapse of major segments of continental crust (an idea that goes back to Descartes), somewhat analogous to modern landslips but on a far larger scale (Rudwick 2001, 2005, pp. 150–158). On the other hand, in 1791 the French naturalist Déodat de Dolomieu (1750–1801) attributed them instead to what would now be called a megatsunami, analogous to the devastating tsunami that had followed the notorious Lisbon earthquake of

1755, but again on a far larger scale (Rudwick 2005, pp. 317–324).

After the turn of the century, the great Parisian naturalist Georges Cuvier (1769–1832) wavered between these two alternative explanations. However, he was most concerned simply to demonstrate that the event was real, very widespread and probably even global; and that it was capable of having wiped out the megafauna of apparently extinct mammals, such as mammoths and mastodons, that he was reconstructing from their fossil bones (Rudwick 1997). This would prove that extinction itself was a natural process and not just due to human agency. Far from privileging the biblical story of Noah's Flood, Cuvier treated it as just one of several accounts, all of them more or less obscure, from various ancient cultures ranging as far away as China. And far from being a biblical literalist, Cuvier cited Johann Eichhorn (1752–1827), the great Göttingen orientalist and biblical scholar, to help put a reliable date on the Flood; he used the biblical story simply as a surrogate for the contemporary records of ancient Egypt, which could not yet be deciphered. Only in the Anglophone world was Cuvier misrepresented, when the editions of his work published by the Edinburgh geologist Robert Jameson (1774–1854) asserted that the Frenchman's primary objective was to bolster the authority of the Bible (Rudwick 2005, pp. 557–571, 585–598).

Biblical Flood and 'geological deluge'

Most geologists accepted the plausibility of this idea of a sudden catastrophic event in recent geohistory. However, many of them disagreed with de Luc, Dolomieu and Cuvier on its date. They argued that it was too ancient to be equated with the biblical event, or they simply ignored the question of its date in relation to human history. In effect, the 'geological deluge' (as it came to be called) was distinguished more and more sharply from the biblical Flood: the question of the (geo)historical reality of the physical event was increasingly separated from the question of its chronological date. (This analytical distinction between biblical 'Flood' and geological 'deluge' is not difficult to detect from the contexts in which the terms were used, although the writers themselves did not use the terms consistently in this way, and of course they were referring to the putative event in several different languages.)

For example, towards the end of the Napoleonic wars in 1815, the great Prussian geologist Leopold von Buch (1774–1853) tackled the most puzzling of all the 'diluvial' problems, namely the huge erratic blocks scattered erratically around the Alps



Fig. 1. The Pierre à Bot, a huge granite erratic block stranded on the forested slopes of the Jura range above Neuchâtel, about 100 km from its bedrock source near Mont Blanc, as sketched in 1820 by Henry De la Beche (1796–1855). In 1815 Leopold von Buch had used this erratic as his prime example of evidence for the geohistorical reality of an exceptional aqueous catastrophe, subsequently termed the ‘geological deluge’, in the geohistorically recent past (reproduced with permission of the National Museum of Wales).

and across northern Europe, far from where the relevant rocks cropped out *in situ*. Von Buch’s prime example was the Pierre à Bot high above Neuchâtel (Fig. 1). How had this block of granite the size of a small house been moved over 100 km from its unquestioned source in the Mont Blanc massif, down the upper Rhône valley, across Lac Léman (the Lake of Geneva), over the low hills of the Pays de Vaud, and right up onto the slopes of the Jura range? Von Buch was utterly perplexed; but in 1818 a terrible disaster in the Val de Bagnes (coincidentally, not far from the source area of the granite of the Pierre à Bot) gave him a possible *cause actuelle* for it. At some point in the unrecorded distant past there might have been a similar but far larger and more catastrophic mudslide or subaerial turbidity current, flowing all the way from the high Alps to the Jura. Around the same time as von Buch’s work, James Hall (1761–1832), the younger friend of the deceased James Hutton (1726–1797), described surfaces of scratched bedrock and other strange linear features around Edinburgh. He scaled up the *cause actuelle* of known tsunamis, and followed Dolomieu in suggesting, in a highly un-Huttonian manner, that the enigmatic features were due to a mega-tsunami. Neither von Buch nor Hall referred to the likely dates of these catastrophic events: the reality of the putative geological deluge was dissociated from any explicit reference to the biblical Flood (Rudwick 2005, pp. 573–584).

However, other geologists did follow de Luc, Dolomieu and Cuvier in equating these two events, and in the Anglophone world this coincidence was

used by some of them to support the trustworthiness of the Bible as a whole. For example, the Oxford geologist William Buckland (1784–1856) mapped ‘diluvial’ gravels across the English Midlands, which contained distinctive erratic pebbles that had been carried southwards right over a watershed, in a way that seemed inexplicable in relation to the present rivers. He therefore inferred that the gravels were due to some kind of exceptionally violent ‘diluvial current’ or mega-tsunami. Soon afterwards fossil bones were discovered in Kirkdale Cave in northern England. In a careful analysis of this site, Buckland reconstructed the small cave as a former den of extinct ‘antediluvial’ hyaenas. His friend and colleague William Conybeare (1787–1857) caricatured this by showing Buckland time-travelling back into antediluvial geohistory, ‘bursting the limits of time’ in just the way that Cuvier had famously advocated for geology as a whole (Fig. 2). Buckland then turned his attention to many other caves, in Britain, France and Bavaria, that were already known to contain fossil bones; and he used them all as further evidence for the reality and very wide impact of the ‘geological deluge’ (Rudwick 2005, pp. 600–620).

The dissociation of Flood and deluge

In the religiously conservative environment of the University of Oxford, the intellectual centre of the Church of England, Buckland the Anglican cleric argued that the geological deluge was none other than the biblical Flood, although he, like de Luc and other earlier writers, had to interpret the story of Noah in a far from literalistic manner. However, this was not a battle between ‘science’ and ‘religion’. One of Buckland’s most forceful critics in Britain was the Scottish naturalist and Presbyterian (Calvinist) cleric John Fleming (1785–1857), who rejected the reality of the geological deluge altogether, and doubted if the biblical event had left any physical traces, at least in his own part of the world (Burns 2007; Rudwick 2008, 82–86). Fleming denounced Buckland’s diluvial theory, calling it (quoting Francis Bacon) ‘Philosophia phantastica, religio haeretica’. On the other hand, many other critics of Buckland, such as the London geologist William Fitton (1780–1861) and others elsewhere in Europe, accepted the reality of the geological deluge but inferred that it was much too ancient to be equated with the biblical Flood.

During the 1820s and 1830s this dissociation between geological deluge and biblical Flood became more marked among geologists of all European nations. In England a decisive shift in expert opinion came when Adam Sedgwick



Fig. 2. William Buckland crawling into Kirkdale Cave in 1821 with the light of science in hand, and finding that the extinct cave hyaenas, whose habits he had reconstructed from their fossil bones, were alive and well: a caricature by his friend William Conybeare. This putative ‘antediluvial’ hyaena den was an important part of Buckland’s evidence for the reality of a ‘geological deluge’, which he, unlike many other geologists, equated with the biblical Flood (reproduced from the author’s collection).

(1785–1873), Buckland’s counterpart at Cambridge and like him an Anglican cleric, defected from Buckland’s side to Fitton’s. Sedgwick concluded that the diluvial deposits dated from more than one period in geohistory, and that all of them were probably too ancient to be identified with the biblical Flood. He found the extensive fieldwork of Léonce Élie de Beaumont (1798–1874), a Parisian geologist of a younger generation than Cuvier’s, particularly persuasive. Élie de Beaumont distinguished older from newer diluvial deposits, and attributed them to distinct and successive ‘*époques de soulèvement*’. According to his tectonic theory, which he had developed from von Buch’s earlier ideas, these putative episodes of relatively sudden crustal buckling had elevated new mountain ranges and generated mega-tsunamis at distant intervals throughout geohistory (Rudwick 2008, 113–114, 129–133, 333–336). Whether the most recent upheaval had been recent enough for its mega-tsunami to have been recorded by early

human societies was left unresolved; but it seemed certain that few of the ‘diluvial’ deposits, and perhaps none of them, had anything to do with the biblical event.

During these same decades, fieldwork by many other geologists showed that the diluvial effects were astonishingly widespread across northern Europe and all round the Alps (Fig. 3). In St. Petersburg, for example, Gregor Kirilovitch, count Razumovsky (d. 1837; the brother of the Razumovsky who was Beethoven’s patron in Vienna), described erratics that had crossed the Gulf of Finland onto the plains of northern Russia. Cuvier’s Parisian colleague Alexandre Brongniart (1770–1847), while visiting Scandinavia primarily in search of trilobites, mapped eskers and erratics trending southwards across southern Sweden. Johann Hausmann (1782–1859) of Göttingen described other erratics of Scandinavian origin that had crossed the Baltic onto the north German plain, some of them even reaching the Netherlands.

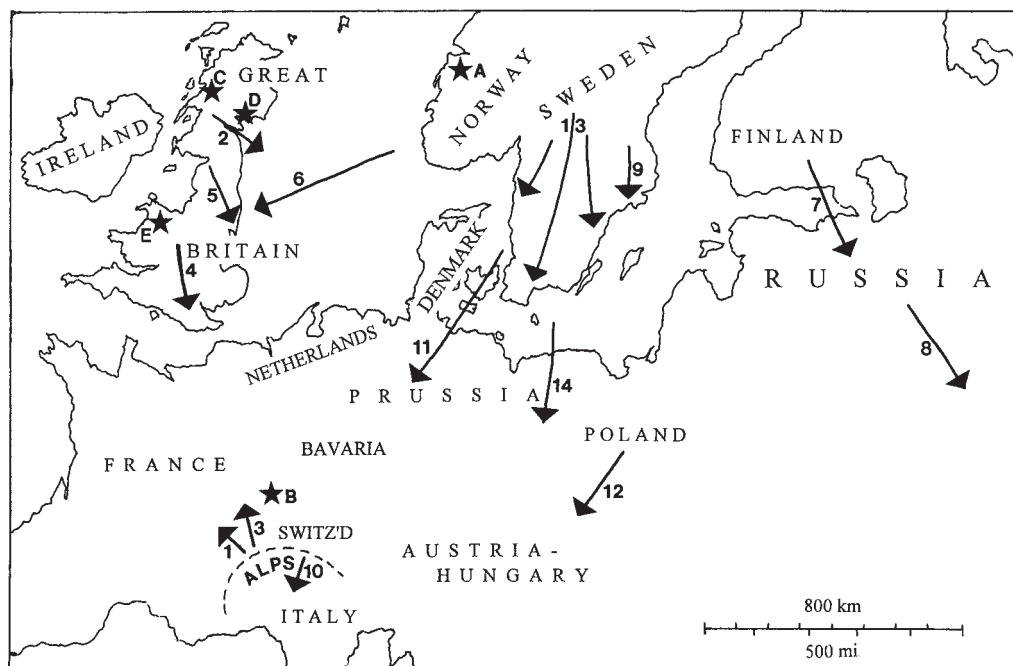


Fig. 3. A map of Europe, showing the inferred tracks of various 'diluvial currents' that had carried erratic blocks and other debris far from their source areas, as reported mostly in the 1820s and 1830s. They include those described by Brongniart (9), von Buch (3, 10), Buckland (4–6), Hall (2), Hausmann (11), Pusch (12), Razumovsky (7) and Sefström (13). Many were later reinterpreted as the result of glacial action in a geohistorically recent but prehistoric Ice Age; the asterisks A–E denote areas in which the traces of former valley glaciers were detected, which were important in this transformation of the diluvial into the glacial theory.

Buckland noted that yet others had crossed the North Sea onto the east coast of England. Further south, von Buch found trails of erratics on the southern flanks of the Alps to match those he had already traced on the north. The mining geologist Georg Pusch (1790–1846) of Warsaw mapped diluvial deposits and erratics extending from his home city southwards towards Krakow, and attributed them to 'a colossal flood penetrating with immense velocity [mit ungeheurer Geschwindigkeit]'. And another mining geologist, Nils Sefström (1787–1845) of Falun, mapped in detail the scratched bedrock surfaces all across southern Sweden, and attributed them similarly to a huge and violent south-trending 'petridelauniska floden' or flood of little stones, apparently long before any humans were around.

Meanwhile, similar reports from beyond Europe suggested that the diluvial currents were even more widespread and possibly worldwide. For example, John Bigsby (1792–1881), a British physician attached to the commission surveying the disputed frontier between the USA and Canada, reported vast spreads of erratics around the shores of Lake Huron. Also, Buckland, profiting from British

expeditions to Burma and Alaska, argued that the megafauna of extinct mammals that Cuvier had first reconstructed had flourished in antediluvial times in all latitudes from the tropics to the Arctic (Rudwick 2008, 185–189, 196–198, 501–505). Apart from Buckland, however, most of the geologists mentioned above inferred, explicitly or at least implicitly, that the enigmatic diluvial event was too far in the deep past of geohistory to have left any record in human history. The consequent dissociation between geological deluge and biblical Flood was generally amicable. To repeat the point: it was not an argument between 'science' and 'religion' or between religious believers and sceptics.

The transcendence of literalism

The role of biblical studies in the nineteenth-century debates about the relation between religion and the natural sciences has been woefully neglected. In fact, scholarly methods of biblical interpretation, based on greatly improved knowledge of the relevant ancient languages and cultures, were already flourishing, particularly in some of

the German universities, where such work built on earlier research by scholars such as Eichhorn and the great Johann Michaelis (1717–1791) at Göttingen (Löwenbrück 1986; Sheehan 2005). By the early nineteenth century, educated people in most European countries, including those who would now be called ‘scientists’, were coming to recognize that biblical literalism was no longer tenable, and that it had not been characteristic of Christian thinking in the earlier history of the Church (Frei 1974; Rogerson 1984; Harrison 1998). In the case of geology, the recorded comments of geologists show that even in Britain at least some of them were aware of this. Conybeare, for example, who did distinguished work in both geology and theology, deplored his compatriots’ ignorance of the work of German biblical critics such as Eichhorn, and insisted that ‘the Bible is exclusively the history of the dealings of God with men’, and that it should not be misused as a quarry for scientific data (Rudwick 2008, 423–427).

This newly historical understanding of all ancient texts, including but not only biblical texts, allowed the putative universality of the biblical Flood, for example, to be reinterpreted in terms of the likely perceptions of ancient cultures. The biblical scholars concluded that the Flood story probably referred to a catastrophic event confined to some limited area (perhaps Mesopotamia, the traditional site of Noah’s embarkation in his Ark) that had already been settled by people of an early literate culture: it might indeed have been ‘universal’, but only in terms of the world as they experienced it. Any physical traces of the historical reality of the Flood would therefore have to be sought in that part of the world, not in Europe. The assimilation of scholarly textual criticism therefore led to the geological marginalization of the biblical Flood, which was now assumed to have been confined to the limited region then settled by the earliest human societies. Even Buckland retracted his earlier emphatic claim that the diluvial features in northern Europe had been due to the effects of the biblical Flood. In effect, the biblical Flood was demythologized (to use Bultmann’s classic term) into a localized inundation in early human history.

In contrast, the much earlier and apparently pre-human ‘geological deluge’, attributed to some kind of massive tsunami-like event, was taken to have spread diluvial gravel and erratic blocks around Europe and North America, and perhaps even globally, while also causing the mass extinction of Cuvier’s fossil megafauna (most of it, in present-day terms, of Pleistocene age). However, this natural explanation was soon transcended, and in part superseded, by the concept of a geohistorically recent *Eiszeit* or Ice Age. The glacial theory was proposed most sensationally in 1837 by the

palaeontologist Louis Agassiz (1807–1873) of Neuchâtel, but more soundly by other Swiss geologists such as Ignace Venetz (1788–1859), Jean de Charpentier (1786–1855) and Bernhard Studer (1794–1887). It eventually provided a far more satisfactory explanation for most of the diluvial features, and particularly for erratic blocks. Buckland was an early and enthusiastic convert to glacialism, and during the 1840s most other geologists came to agree that the putative geological deluge had in reality been some kind of Ice Age, though few of them adopted Agassiz’s extreme ‘Snowball Earth’ version of it (Rudwick 2008, 508–539).

In retrospect, it is arguable that the relation between the Flood story in Genesis and the findings of the new science of geology would never have become such a focus of debate had it not been for the deeply human-historical character of Judaeo-Christian religion and the newly geohistorical orientation of geology. The novelty of the latter deserves some emphasis. Traditionally, the Earth sciences had not been historical in outlook: in earlier periods they had emulated either the ahistorical classificatory goals of other branches of natural history, or the ahistorical causal goals of other branches of natural philosophy. As mentioned above, it was only in the late eighteenth century that the Earth sciences took a historicizing turn, when methods, models and metaphors from the study of human history were knowingly transposed into the natural world. Half a century later, this geohistorical practice had become so firmly embedded in the science of geology that in 1837 the Cambridge polymath William Whewell (1794–1866), then acting as president of the Geological Society, coined a new name to denote this new kind of natural science. In Whewell’s mapping of all the sciences, geology was ‘palaeiological’ because it combined ‘palaeontology’, literally the study of past entities of all kinds (not only fossils), with ‘aetiology’, the study of their causal origins. This was the kind of historicized natural science that the young Charles Darwin (1809–1882) (‘I, the geologist’, as he then called himself) imbibed at that time, and soon set about extending from the inorganic world into the world of organisms (Rudwick 2008, 489–493, 546–548).

Scientific and religious meanings

The story of biblical Flood and geological deluge, summarized here very briefly, is all too readily turned into grist for any number of anti-religious mills and tendentious ‘conflict’ narratives. However, there are other interpretations available. The gradual and generally peaceable dissociation between biblical

Flood and geological deluge is a good example of the general human learning process by which the appropriate fields of application of diverse claims to reliable knowledge come to be recognized and differentiated.

The immediate source for the historicizing of nature in the new science of geology was the archaeological work that, for example, reconstructed Pompeii and Herculaneum from their excavated ruins (Rudwick 2005, pp. 181–194). However, a more profound inspiration, as mentioned above, was the overarching Judaeo-Christian sense of human history as complex, unrepeated and, above all, contingent. Transposed from culture into nature, this basic religious sense, far from ‘retarding the progress of science’ as crusading atheistic fundamentalists often claim, lies at the very roots of the modern Earth sciences. However, it remains an open question whether a modern historicized and demythologized interpretation of the ancient story of Noah’s Flood is capable of carrying the imaginative religious meaning that it embodied for earlier ages. This we can at least dimly re-experience, for example in Britten’s powerful musical reworking of the mediaeval play *Noyes Fludde* (1958), just as Haydn’s *Die Schöpfung* (1798) can allow us to re-experience the earlier imaginative impact of the creation story. Perhaps the whole effort to match the Genesis stories with the science of geology was radically misconceived, in that the newer historically informed biblical interpretation, no less than the literalistic one it sought to replace, failed to put the question of religious meaning centre stage.

Meanwhile, however, and as a properly separate project, glaciologists and Quaternary geologists now find ever richer scientific meaning in the physical features that for a time were fruitfully interpreted as the traces of some kind of ‘geological deluge’. In conclusion, therefore, historians of the relation between geology and religion should not be content either with a conflict model or with a harmony model, either with overlap or with non-overlap between ‘science’ and ‘religion’. In the case summarized here, the diverse human learning processes that in German are rightly called *Wissenschaften* (‘sciences’ in the plural, both natural and human) led in the course of time to the differentiation of properly distinct spheres of enduring meaning, both scientific and religious.

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